

Assignment: Polynomial Regression

Vinay Kusumanchi

Roll No: IMT2023608

GitHub: <https://github.com/kvinayy/ML-Assignment-1>

1 Problem and Approach Overview

Two personalised regression problems were given. Each has 1000 training and 1000 test samples, all features lie in $[-1, 1]$, and only polynomial regression is allowed.

- **var1** (steam turbine): six inputs $x_1, \dots, x_6$, target y is the Net Power Score. Maximum allowed degree is 10.
- **var2** (thermal reservoir): three inputs x_1, x_2, x_3 (spatial offsets), target y is the Thermal Anomaly Score. Maximum allowed degree is 20.

The same pipeline was applied to both problems: (1) load and inspect the data, (2) exploratory analysis, (3) linear baseline, (4) polynomial degree search with cross-validation, (5) feature-subset search, (6) overfitting analysis, (7) Ridge/Lasso regularisation, (8) final model selection, (9) diagnostics, and (10) test prediction with format validation. Test labels are hidden, so every selection decision rests on cross-validation over the training data only.

2 Experimental Setup

All code is in Python 3.13 using NumPy, pandas, scikit-learn, SciPy and Matplotlib.

Model pipeline. Every polynomial model is a scikit-learn Pipeline:

```
PolynomialFeatures(degree, include_bias=False) → StandardScaler → estimator,
```

where the estimator is `LinearRegression`, `Ridge` or `Lasso`. Scaling after feature expansion keeps high-order terms on comparable scales, which matters for conditioning and for the penalty in Ridge/Lasso. Because the scaler sits inside the pipeline, it is re-fit on the training part of each fold, so no information leaks from validation folds.

Validation protocol. All models were compared with 5-fold KFold (shuffled, `random_state=42`). The same folds were used for every model, so scores are directly comparable. For each model we recorded mean CV MSE (with its standard deviation), mean CV R^2 , and the mean train MSE and R^2 across folds.

Number of terms. For d features and degree k , the expansion has $\binom{d+k}{k} - 1$ terms. For var1 this grows from 6 (degree 1) to 209 (degree 4), 461 (degree 5), 923 (degree 6) and 8007 (degree 10). For var2 it grows from 3 (degree 1) to 164 (degree 8), 363 (degree 11) and 1770 (degree 20). Each CV training fold has 800 samples, so the expansion has more terms than samples from degree 6 onwards for var1.

3 Data Inspection and Exploratory Analysis

Integrity checks. Training sets have shape (1000, 7) for var1 and (1000, 4) for var2. Test sets have 6 and 3 feature columns. There are no missing values, all columns are `float64`, train and test column names match, and the training sets contain no duplicate rows. The test sets contain 4 (var1) and 10 (var2) duplicate rows, which were left as they are, since one prediction per test row is required.

Target statistics. var1: y has mean 0.86, std 3.39, range $[-11.43, 14.35]$. var2: mean 2.16, std 6.81, range $[-29.99, 29.84]$.

Train/test comparison. In var2, train and test feature statistics are very close. In var1, the test features have a visibly larger spread (std ≈ 0.81 – 0.82 vs. ≈ 0.71 – 0.73 in train). So test points sit closer to the boundaries ± 1 , where high-degree polynomials are least reliable. This favoured choosing a model that is stable near the edges rather than one that only fits the interior well.

Plots. Correlation matrices, feature-vs-target scatter plots and feature histograms were generated for both problems. The scatter plots show no single dominant linear trend in y against any feature, which is the first sign that a linear model will underfit.

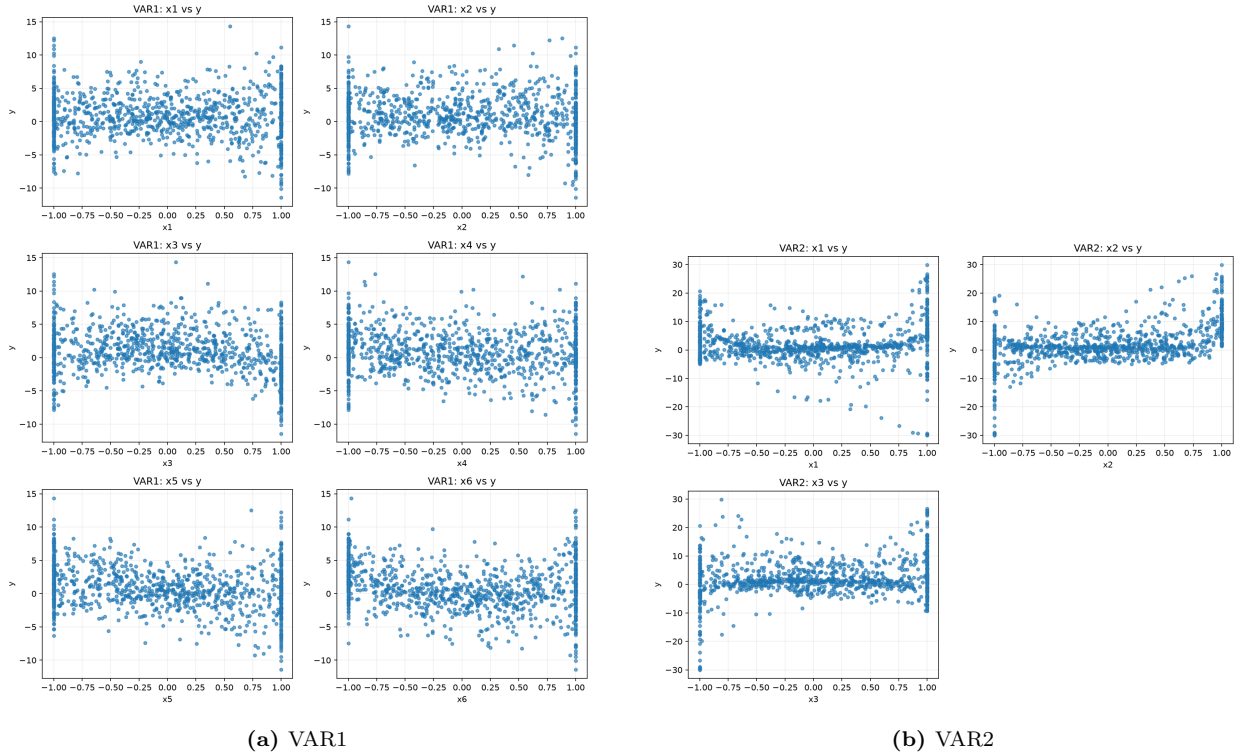


Figure 1: Feature vs. target scatter plots on the training data.

4 Linear Baseline

A scaled linear regression gave:

Problem	CV MSE	CV R^2	Train MSE	Train R^2
var1	10.313	0.097	10.054	0.126
var2	32.488	0.296	32.187	0.305

Train and CV errors are almost equal, so the linear model is *underfitting* (high bias), not overfitting. This justifies moving to higher-degree polynomials.

5 Polynomial Degree Search

Plain polynomial regression (all features, no penalty) was evaluated for every degree from 1 to 10 (var1) and 1 to 20 (var2).

Degree	var1		var2	
	CV MSE	CV R^2	CV MSE	CV R^2
1	10.3125	0.0973	32.4883	0.2957
2	3.4960	0.6926	20.8794	0.5478
3	1.0800	0.9042	9.6334	0.7909
4	0.8214	0.9270	3.5727	0.9226
5	1.6584	0.8545	1.3767	0.9700
6	80.7414	-6.2163	0.5956	0.9869
7	9.0295	0.2022	0.3307	0.9927
8	6.4353	0.4334	0.2888	0.9936
9	5.6946	0.4968	0.3282	0.9927
10	6.0129	0.4730	0.3722	0.9918

Table 1: Plain polynomial CV results (5-fold). For var2, degrees 11–20 continue to worsen: CV MSE is 0.463 at degree 11, 1.014 at 12, 6.481 at 13, 48.4 at 14, and reaches 592.1 at degree 20.

var1. CV MSE falls steeply up to degree 4 (0.821, $R^2 = 0.927$) and then rises. At degree 6 it collapses (MSE 80.7). **var2.** CV MSE falls until degree 8 (0.289, $R^2 = 0.994$) and then increases, becoming catastrophic from degree 13 onward.

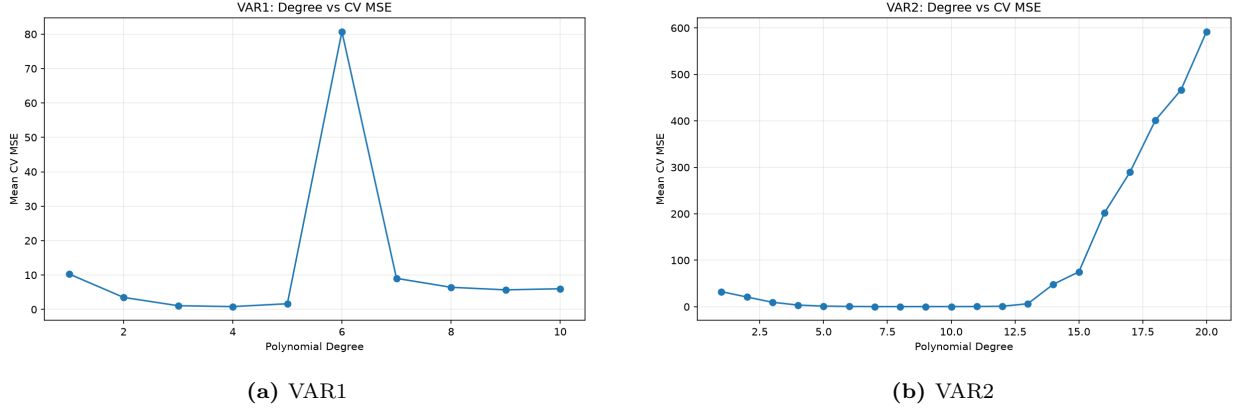


Figure 2: Polynomial degree vs. mean CV MSE.

6 Overfitting Analysis

For each degree, the gap between CV and training error was tabulated (MSE gap = CV MSE – train MSE, R^2 gap = train R^2 – CV R^2).

var1. The gap is small up to degree 3 (MSE gap ≈ 0.25), grows at degree 4 (0.45) and degree 5 (1.54), and explodes from degree 6. For degrees 6–10 the train MSE is identical (0.000861, train $R^2 = 0.9999$) while CV MSE is 5.7–80.7. This is consistent with the models interpolating the training folds: from degree 6 there are 923 or more terms against 800 training samples per fold, so the fit can pass through every training point and generalises poorly.

var2. Train MSE keeps decreasing all the way to degree 20 (0.022) while CV MSE is minimal at degree 8 and then grows (gap 0.12 at degree 8, 0.93 at degree 12, 592 at degree 20). The classical U-shaped validation curve is clear: low degrees underfit, very high degrees overfit.

7 Feature-Subset Search

To check whether any inputs were irrelevant, an exhaustive search over all non-empty feature subsets was run, with polynomial degrees 1–5 for var1 (63 subsets, 315 CV evaluations) and 1–8 for var2 (7 subsets, 56 CV evaluations).

- **var1:** the best subset is all six features at degree 4 (CV MSE 0.821). The best 5-feature subset (x_1, x_2, x_3, x_5, x_6 , degree 4) is clearly worse (MSE 2.000), so every feature, including x_4 , carries signal.
- **var2:** the best subset is all three features at degree 8 (CV MSE 0.289). The best 2-feature subset (x_1, x_2) only reaches MSE 17.19, so the vertical offset x_3 is essential.

Hence no feature was dropped for either problem.

8 Ridge and Lasso Regularisation

Plain least squares degrades quickly past its optimal degree, so ℓ_2 (Ridge) and ℓ_1 (Lasso) penalties were tried to allow slightly richer polynomials without overfitting. The search used the best feature subsets from the previous step (all features for both problems):

	Degrees	Ridge α	Lasso α
var1	3, 4, 5, 6	0.001, 0.01, 0.1, 0.3, 1, 3, 10	10^{-4} , 3×10^{-4} , 0.001, 0.003, 0.01, 0.03, 0.1
var2	7, 8, 9, 10, 11	same	same

Lasso used `max_iter=100000` and `tol=10-4`. Failed or non-converging configurations were scored as NaN (via `error_score`) so that one unstable setting could not stop the search.

Problem	Model	Degree	α	CV MSE	CV MSE std	CV R^2
var1	Plain polynomial	4	—	0.8214	0.1101	0.9270
var1	Ridge	5	10	0.5196	0.0668	0.9542
var1	Lasso	5	0.01	0.3600	0.0324	0.9683
var2	Plain polynomial	8	—	0.2888	0.0539	0.9936
var2	Ridge	11	1	0.2643	0.0429	0.9942
var2	Lasso	11	0.001	0.2675	0.0417	0.9941

Table 2: Best configuration of each model family (5-fold CV).

For var1 regularisation gives a large gain: CV MSE drops from 0.821 to 0.360 (about 56%), and the CV spread also shrinks. The optimal degree moves from 4 to 5, because the penalty controls the extra terms that plain least squares cannot handle. For var2 the gain is small (0.289 to 0.264), which fits the plain polynomial already being near-optimal there.

9 Final Model Selection

Candidates (best plain polynomial, best Ridge, best Lasso) were ranked by lowest CV MSE, with CV MSE standard deviation and then CV R^2 as tie-breakers. The selected models are:

Problem	Model	Degree	Features	α
var1	Lasso	5	$x_1, \dots, x_6$	0.01
var2	Ridge	11	x_1, x_2, x_3	1.0

The chosen degrees (5 and 11) respect the limits in the problem statement (10 and 20). **Rationale.** For var1, Lasso at degree 5 has the lowest CV MSE (0.360) and the smallest fold-to-fold variation (0.032), so it is both the most accurate and the most stable choice. For var2, Ridge at degree 11 has the lowest CV MSE (0.264); degree 11 is high only because the penalty tames the extra terms, while the plain degree-11 model is clearly worse (0.463). Ridge and Lasso are within noise of each other there, and Ridge won on the stated criterion.

Caveat. The same CV folds were used to tune degree, penalty and model type, so the reported CV scores are slightly optimistic compared with a truly untouched test set. No separate hold-out set was used, since the test labels are hidden.

10 Final Fit and Training Diagnostics

Both final models were fitted only on the complete training sets and then evaluated on the training data for diagnostics.

Problem	Train MSE	Train RMSE	Train R^2	Residual mean	Res. min	Res. max
var1	0.2525	0.5025	0.9781	0.0000	−1.367	1.802
var2	0.1701	0.4124	0.9963	0.0000	−1.292	1.477

Residuals are centred at zero (medians 0.0006 and -0.0025) and bounded within roughly ± 1.8 against target standard deviations of 3.39 and 6.81. Residual-vs-prediction and Q-Q plots were produced for both models.

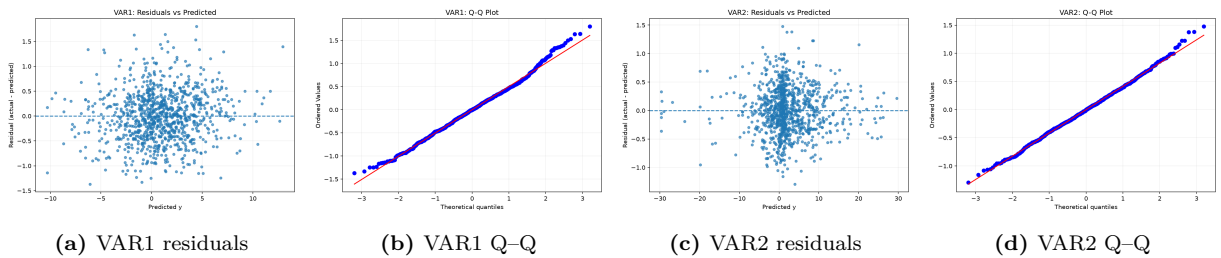


Figure 3: Diagnostics of the final models on the training data.

11 Test Predictions and Submission Validation

Predictions were generated for the 1000 test rows of each problem, using the same feature columns as the final model.

Problem	Rows	Min	Max	Mean
var1	1000	-10.61	16.12	1.09
var2	1000	-29.75	37.07	2.46

The predicted ranges are close to the training target ranges. In var2 the maximum prediction (37.07) exceeds the training maximum (29.84), suggesting some extrapolation at test points near the domain edges, which is a known risk of high-degree polynomials.

The files `IMT2023608_pred_var1.csv` and `IMT2023608_pred_var2.csv` each contain a single column named `y` and no index column. Automated checks confirmed the correct row count, correct column name, no missing values, numeric and finite values, and no accidental index column. Both files passed all checks.

12 Conclusion

For var1, the final model is a degree-5 polynomial in all six features with a Lasso penalty ($\alpha = 0.01$), reaching CV MSE 0.360 and CV R^2 0.968, compared with 10.31 and 0.097 for the linear baseline. For var2, it is a degree-11 polynomial in all three coordinates with a Ridge penalty ($\alpha = 1$), reaching CV MSE 0.264 and CV R^2 0.994, compared with 32.49 and 0.296 for the baseline. In both problems, the degree and the penalty were chosen by cross-validation, not by training error. The experiments show that unpenalised polynomials overfit sharply beyond their optimal degree, while a well-tuned Ridge or Lasso penalty improves generalisation, most clearly for var1.